Computing derivatives using the definition can be quite a lot of work. Over the next few lessons we are going to prove various theorems which make computing derivatives much easier by allowing us to build derivatives of complicated functions out of simpler ones.

We have proved previously that if n is a positive integer, then

$$\frac{d}{dx}x^n = nx^{n-1}. \tag{1}$$

The formula in Equation 1 is called the *Power Rule*.

1. Use the Power Rule to compute the derivative of each of the following functions.

   (a) $g(t) = t^{14}$

   (b) $f(x) = x^{37}$

   (c) $h(x) = x^{1000}$

   Though we do not yet have the technology to prove it, the Power Rule actually holds for arbitrary *real* exponents; that is, if α is any real number, then

   $$\frac{d}{dx}x^\alpha = \alpha x^{\alpha-1}$$

   wherever $x^{\alpha-1}$ is defined.

   Examples

   (a) $\frac{d}{dx}x^{-2} = -2x^{-2-1} = -2x^{-3} = -\frac{2}{x^3};\ x > 0.$

   (b) $\frac{d}{dx}x^{\frac{1}{2}} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}};\ x > 0.$

   (c) $\frac{d}{dx}x^\pi = \pi x^{\pi-1};\ x \geq 0.$

2. Prove the *Constant Multiple Rule*: If c is a constant and f is a differentiable function, then

$$\frac{d}{dx}(cf(x)) = c\frac{d}{dx}f(x).$$

(*Hint*: Let $F(x) = cf(x)$. Write out the definition of $\frac{d}{dx}F(x)$ then substitute.)

3. Compute the derivative of each of the following functions.

(a) $f(x) = 3x^4$

(b) $g(x) = -x$

4. Prove the *Sum Rule*: If f and g are both differentiable, then

$$\frac{d}{dx}(f(x)+g(x)) = \frac{d}{dx}f(x) + \frac{d}{dx}g(x).$$

(*Hint*: Let $F(x) = f(x)+g(x)$. Write out the definition of $\frac{d}{dx}F(x)$ then substitute.)

5. The sum rule can be extended to any number of functions:
$(f_1 + f_2 + \cdots + f_n)' = f_1' + f_2' + \cdots + f_n'$. Use this extended Sum Rule together with the constant multiple to compute the derivative of a polynomial

$$f(x) = \sum_{k=0}^{n} c_k x^k = c_0 + c_1 x + c_2 x^2 + \cdots + c_{n-1} x^{n-1} + c_n x^n.$$

Justify each step.

6. Compute the derivative of the polynomial $3x^5 + 2x^4 + x^3 + 14x^2 + 6x + 5$.

7. Use the Constant Multiple Rule and the Sum Rule to prove the *Difference Rule*

$$\frac{d}{dx}(f(x) - g(x)) = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$$

8. Compute the derivative of $f(x) = x^6 - x^4$.